VASANTDADA PATIL PRATISHTHAN'S
COLLEGE OF ENGINEERING AND VISUAL ARTS
(An Autonomous Institute)
NAAC 'A' Grade Accredited, ISO 9001:2015 Certified

Department of Information Technology
NBA Accredited (Dated 01/07/2024 to 30/06/2027)

A

Major Project Report

on

Your Title of Project

Submitted in partial fulfillment of the requirement for the degree of

Bachelor of Engineering

Information Technology Semester – VII

By

Student 1 Name Roll No

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Under the guidance of

Prof. Your Project Guide Name

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University of Mumbai

2026–27



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CERTIFICATE

This is to certify that the project entitled **"Write your project title here"** is bonafide work of Student 1 Name (Roll No), Student 2 Name (Roll No), Student31 Name (Roll No), Student 4 Name (Roll No), submitted to University of Mumbai in partial fulfillment of requirement for the award of degree of "Bachelor of Engineering in Information Technology Semester VII.

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Project Report Approval for B. E.

This project report entitled **"Insert your project title here"** byStudent 1 Name (Roll No), Student 2 Name (Roll No),Student 3 Name (Roll No), Student 4 Name (Roll No), is approved for the degree of Bachelor of Engineering in Information Technology.

Examiners

1. _____

2. _____

Date:

Place:

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DECLARATION

We declare that this written thesis submission represents the work done based on our and/or others' ideas with adequately cited and referenced the original sources. We also declare that we have adhered to all principles of intellectual property, academic honesty, and integrity as we have not misinterpreted, fabricated, or falsified any idea/data/fact/source or original work/matter in our submission.

We understand that any violation of the above will be cause for disciplinary action by the college and may evoke penal action from the sources which have not been properly cited or from whom proper permission is not sought.

1. Student 1 Name Roll No _____
2. Student 2 Name Roll No _____
3. Student 3 Name Roll No _____
4. Student 4 Name Roll No _____

Date:

Place:

ACKNOWLEDGEMENT

We the students from Vasantdada Patil Pratishthan's College of Engineering and Visual Arts, Mumbai of Final Year Engineering in the department of 'Information Technology' have great pleasure in presenting our efforts of developing the project named as **"Insert your project title"**.

We would like to express our heartfelt gratitude to **Dr. Pradip Mane**, Head of Department, for providing us with inspiration, valuable guidance, and timely suggestions throughout our Mini project journey.

Our immense appreciation also goes to our esteemed Guide **Prof. Your Guide Name** for his unwavering support and the opportunity he provided us to undertake this challenging project have been truly invaluable. This experience not only allowed us to conduct extensive project work but also exposed us to a wealth of new knowledge for which we are genuinely thankful. Also, we would also like to express our gratitude to the Major Project Coordinator, **Prof Priyanka Manke** for her insightful inputs.

1. Student 1 Name Roll No

2. Student 2 Name Roll No

3. Student 3 Name Roll No

4. Student 4 Name Roll No

Abstract

The Abstract serves as a condensed version of your entire project. According to the syllabus, it must reflect the process of identifying a need and converting it into a problem statement. It acts as a standalone summary that allows a reader to understand the essence of your Web-Based ML project without reading the full document.

Abstract Length Guidelines For a Mini Project of this scale, the abstract should typically be between 250 to 300 words. It should fit on a single page and be written as a single, continuous paragraph or a few short, dense paragraphs.

Detailed Guidelines for Writing the Abstract To meet the assessment criteria for clarity and societal impact, follow this structural flow:

1. Context and Motivation (2–3 sentences) Start with the domain (Web + ML). State the societal or research need identified during your survey.

Example: ”With the rise of digital health, there is a significant need for accessible diagnostic tools for rural populations...”

2. Problem Statement (1–2 sentences)

Clearly define the technical problem you are solving.

Mention the specific gap identified in existing systems.

3. Proposed Solution and Methodology (3–4 sentences)

Identify the specific ML algorithms and web technologies used.

Briefly mention the data source and the ”best possible solution” selected.

Example: ”This project proposes a Flask-based web application integrated with a Random Forest model trained on the XYZ dataset...”

4. Results and Validation (2–3 sentences)

Highlight the ”full functioning of the working model”.

Mention key performance metrics (accuracy, efficiency) obtained through sim-

ulations or testing.

5. Impact and Conclusion (1–2 sentences)

State how the project achieves ”sustainable development” or maps to an SDG.

Summarize the cost-effectiveness and societal value of the final product.

Abstract ”Checklist” for Students No Citations: Do not include references (e.g., [1]) in the abstract. No Figures/Tables: Keep it strictly text-based.

Past Tense: Since the abstract summarizes work already done, use the past tense (e.g., ”The model was trained...” or ”The survey was conducted...”).

Keywords: Include 4–6 keywords at the bottom (e.g., Machine Learning, Web Development, SDG 3, Predictive Modeling).

Keywords: AI, Machine Learning, (include atleast five keywords)

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Acronyms

ML	Machine Learning
DL	Deep Learning
SVM	Support Vector Machine
DT	Decesion Tree
RF	Random Forest
CNN	Convolutional Neural Networks
NB	Naive Bayes
LR	Logistic regression

Replace the aabove acronyms with acronymsused in your project

Chapter 1

Introduction

This project emphasizes the application of engineering fundamentals to solve real-world challenges. The primary goal of this initiative is to leverage the synergy between Machine Learning and Web Development to address specific societal or research needs identified through an extensive survey. By converting these needs into a well-defined technical problem, the project strives to develop a functional working model that is not only innovative but also aligned with Sustainable Development Goals (SDGs). This introductory chapter provides the foundational context for the study, detailing the background, motivation, and technical objectives that guided the development and implementation of the proposed solution.

1.1 Background

Briefly describe the evolution of Web Development and its transition from static pages to intelligent, data-driven platforms.

Introduce Machine Learning as the core logic that enables the system to learn from data and make predictions or decisions.

Explain the synergy between the two: why a web interface is the best delivery mechanism for making an ML model accessible to end-users.

That is clustering of the nearest approximate value. the 42% who reported no physical or chronic conditions as shown in Figure 1.1. public health concerns [2].

1.2 Motivation

Here, you explain the ”Why” behind your project choice. Detail the specific societal or research needs you discovered during your initial survey.

Discuss the current challenges or pain points faced by the target audience.

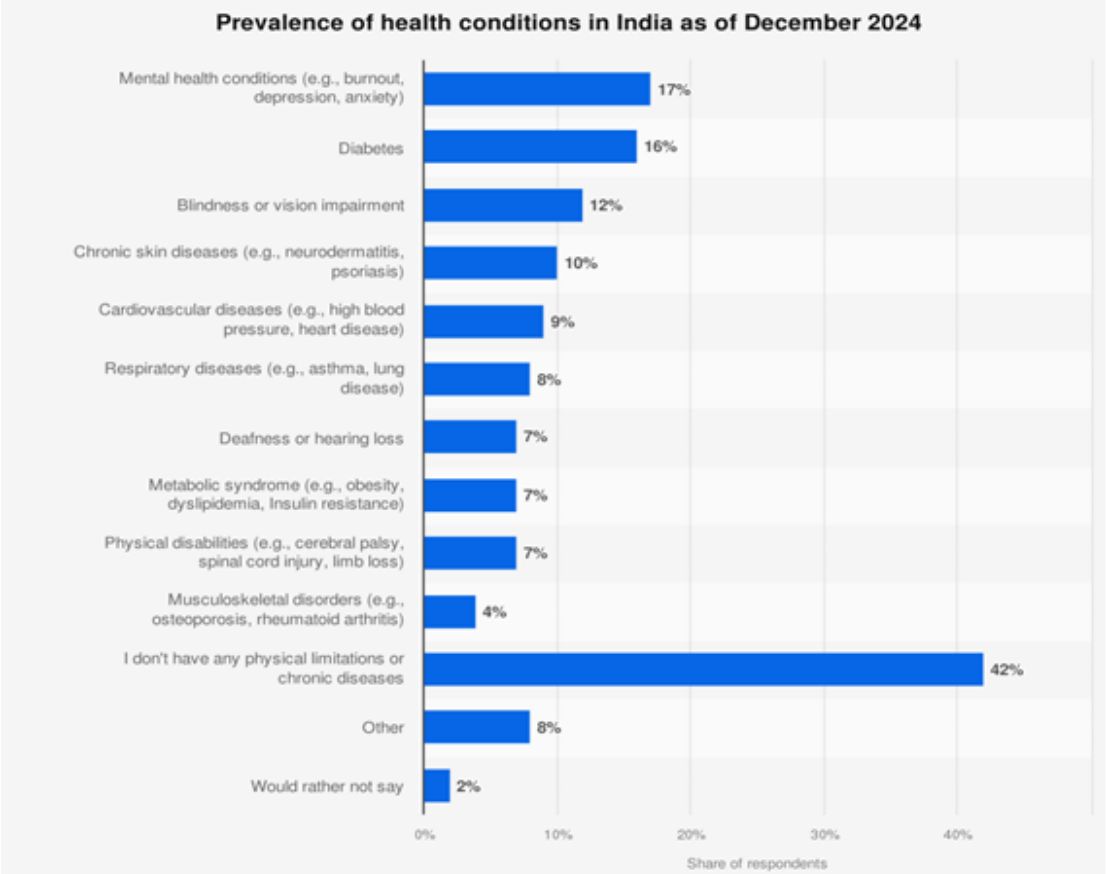


Figure 1.1: Prevalence of health conditions in India as of December 2024 [2]

Mention your desire to apply engineering fundamentals to solve a real-world problem rather than a purely theoretical one.

- Time taken to hire a person.
- Confusion of selecting whom to select.
- To improve the tedious process.

1.3 Problem Definition

This is a precise, technical translation of the motivation. Statement of Need: Summarize the gap identified in Section .

Technical Problem: Convert that need into a technical challenge (e.g., ”Designing a robust CNN-based web application to detect crop diseases with an accuracy of over 90% in real-time”).

Scope: Explicitly state what the project will cover and what it will not, ensuring the project is feasible within the semester timeline.

1.4 Aim and Objectives of the project

1.4.1 Aim

Our aim is to help the HR round go more smoothly and efficiently so that a person's skill can be predicted beforehand without wasting much of the time of the HR. also to sort the resumes according to the requirements of the organizations. And to take the current CV ranking system to an advanced level.

1.4.2 Objectives

Specific goals, including the development of a functional working model. List the "What" of your project in measurable terms.

- Identify and pre-process a domain-specific dataset.
- TDesign and train a specific Machine Learning algorithm.
- Develop a functional working model with a user-friendly web interface.
- Evaluate the model using standard metrics like accuracy, precision, or cost-effectiveness.

1.5 Scope of the Project

IT sector has largest recruiting sector. Our proposed system is initially concerned with IT companies and to recruit the right candidate with perfect skills and technical skills. IT recruitment is in bulk so to reduce the manual work and make sure the intake is of the right candidate. It is also applicable for different jobs and profession.

1.6 Sustainable Development Goal (SDG) Mapping

This is a mandatory requirement for analyzing the impact of your solution. Identify the Goal: Specify the exact UN SDG number and title (e.g., Goal 3: Good Health and Well-being).

Impact Analysis: Provide a detailed explanation of how your specific ML solution contributes to this goal.

Environmental Context: Briefly discuss the sustainability of your solution—how it uses resources efficiently or promotes environmental welfare.

1.7 Organization of the Report

The remainder of this report is organized into five subsequent chapters to provide a comprehensive overview of the project's lifecycle. Chapter 2 presents a detailed literature survey of existing systems and identifies the research gap that justifies the need for this work. Chapter 3 outlines the methodology, system architecture, and project planning, including the Gantt chart for implementation. Chapter 4 details the core implementation phase, covering dataset pre-processing, Machine Learning model development, and the integration of the web-based interface. Chapter 5 provides a demonstration of the working prototype along with an analysis of experimental results and performance metrics. Finally, Chapter 6 concludes the report by summarizing the project's achievements, individual contributions, and potential directions for future research in line with the goals of life-long learning.

Chapter 2

Literature Survey

In Chapter 2, students demonstrate the ”quality of survey/need identification” and ”clarity of problem definition based on need” as required by the syllabus. This chapter provides the scholarly grounding for the project.

2.1 Existing Systems

This section is an analysis of current solutions based on the survey results students are required to conduct.

Current Landscape: Describe the popular web applications or software tools currently used to address the identified problem.

Methodology of Current Tools: Explain the technologies or Machine Learning models these existing systems typically employ.

Functionality: Detail the features provided by these systems and how they serve the current user base.

2.2 Comparative Study

This section summarizes various methods found in research to ”inculcate the process of self-learning and research”.

Algorithm Comparison: Create a table comparing different ML algorithms (e.g., SVM vs. Random Forest vs. CNN) used in previous research papers.

Performance Metrics: Compare results from different theoretical or experimental simulations found in the literature, such as accuracy, latency, or resource consumption.

Technical Constraints: Discuss the hardware and software limitations associated with each researched method.

Studies in this field examine various aspects, including data collection, feature extraction, model training, and performance evaluation, with each study presenting unique insights into the detection process. Table 2.1 highlights their datasets, methods, and results for quick comparison.

Table 2.1: Summary of depression detection using machine learning techniques

Ref.	Author(s)	Year	Results	Limitations
[1]	R. Chiong et al.	2021	98% accuracy with Gradient Boosting; Combines lexical and content features for robustness, strong feature mix, handles uneven data well, beats baseline results.	Needs labelled data, Does not work for other languages, focuses only on Twitter, uses default settings, manual feature design.
[2]	M. S. Liaw et al.	2022	82% F1-score; User engagement metrics improve detection, better user behavior insights	Data privacy concerns; Network features not significant, Some text incomplete or unclear

2.3 Research Gap

This is the most critical part of the survey, where students identify why a new solution is required to fulfill a societal or research need.

- Limitations of Existing Systems: Point out specific shortcomings in current solutions, such as high cost, low accuracy, or lack of a user-friendly web interface.
- Unmet Needs: Highlight the specific societal or research requirements that remain unaddressed by the literature surveyed.
- Justification for New Project: Clearly state how the proposed ”Mini Project - 2 B” will fill these gaps using domain-specific skill sets and engineering fundamentals.

Chapter 3

Design and Planning

In Chapter 3, students translate their research into a concrete engineering plan. This chapter directly addresses the syllabus requirements for ”proposing multiple solutions and selecting the best possible solution” and ”demonstrating project management principles”.

3.1 Proposed Solution

This section describes the ”best possible solution” selected for the underline problem.

Core Approach: Detail the specific Machine Learning algorithm chosen (e.g., Random Forest, CNN, or LSTM) and why it is the most suitable for the identified societal problem.

Web Integration Strategy: Explain how the ML model will be served via the web (e.g., using Flask, Django, or FastAPI).

Innovation: Highlight the ”innovativeness in solutions” as per the assessment criteria.

3.2 System Architecture

This provides a visual and technical roadmap of the ”working model”. Our system follows the three tier architecture. First tier consist of GUI, followed by Processing block and the Database.

3.2.1 GUI (Graphical User Interface)

In our project GUI deals with the interface for the user where the user will have to register and then login give the skill test and submit his resume in any format (Pdf, doc, docs, etc.)

3.2.2 Processing block

This is the block where the actual processing of our project is done. This block will connect the database and the GUI .i.e. it acts as a connector as well as communicator which connects the database and helps in transfer of data between the GUI and the database. Its main function will be to take input from the uploaded CV of the candidate.

3.2.3 Databas:

Database tier is used for the data storage. This contains all the data that is need for the processing and string data of the project. The data in this tier is related to the student information gathered from his/her CV.

3.3 Design

Include the UML Diagram as per requirement of your project

3.3.1 Use case Diagram

3.3.2 Activity Diagram/ Sequence Diagram

3.3.3 Data Flow Diagram

3.4 Algorithm

Explain the algorithms used in your project.

3.4.1 K- Means Algorithm

The Lloyd's algorithm, mostly known as k-means algorithm, is used to solve the k-means clustering problem and works as follows. First, decide the number of cluster: Step 1: Initialize the center of the clusters Step 2:.Attribute the closest cluster to each data point Step 3: Set the position of each cluster to the mean of all data points belonging to that cluster Step 4 : Repeat steps 2-3 until convergence

3.5 Component Selection

This section documents the "system selection and cost analysis" required by the project.

Software Selection: Justify the choice of programming languages (e.g., Python), libraries (e.g., Scikit-learn), and web frameworks based on ”engineering fundamentals”.

Hardware Requirements: List the computational resources needed for training the model and hosting the web application.

Cost Analysis: Provide a detailed breakdown of costs (if any) to ensure ”cost effectiveness,” a key assessment criterion.

3.6 Implementation Plan

Weekly activity breakdown using a Gantt, PERT, or CPM chart. Students must demonstrate ”project management principles” by organizing their work into a schedule.

Weekly Activity Breakdown: Each group must provide an ”implementation plan in the form of Gantt/PERT/CPM chart”.

Milestones: Map key activities—such as ”identification of need,” ”building prototype,” and ”testing”—to specific weeks of the semester.

Self-Learning Progress: Align the plan with the ”log book” entries to record weekly work progress for verification by the project guide.

Chapter 4

Implementation and Testing

In Chapter 4, students document the actual ”building of the working prototype” and ”procurement of systems”. This chapter proves that the ”best solution” has been converted into a ”working model” as required by the syllabus. In this chapter, we show how well the proposed model works for identifying various depression severity levels in tweets. //

As per the requirement of your project develop implementation plan and testing. Here sample implementation plan is given for the AI and ML based project

4.1 Data Collection and Pre-processing

This section describes the raw materials of the project—the data. Dataset Identification: Describe the source of the data (e.g., Kaggle, UCI, or primary survey). Data Description: Detail the features, labels, and size of the dataset.

Pre-processing Steps: Explain how the data was cleaned and prepared, such as handling missing values, encoding categorical variables, or feature scaling.

Data Splitting: Mention the ratio used for training and testing to ensure proper ”validation of results”.

4.2 ML Model Development

Here, students detail the ”Machine Learning” core of the project.

Algorithm Selection: Specify the chosen algorithm and justify why it is appropriate for the ”societal/research needs” identified earlier.

Training Process: Describe the environment, hyperparameters, and the iterative process of training the model.

Model Optimization: Detail any techniques used to improve performance, demonstrating the ”effective use of skill sets”.

Table 4.1 shows the hyperparameter values we used to fine-tune and improve the performance of model.

Table 4.1: Hyperparameters

Parameters	**Values**
Optimizer	Adam
Loss	Sparse Categorical Cross-Entropy
Epochs	10
Activation	Softmax (output layer)
Learning rate	0.00005
Batch size	32

4.3 Web Interface Development

This section covers the ”Web-Based” aspect of the project.

Frontend Design: Describe the user interface (UI) components created to allow users to interact with the ML model.

Backend Integration: Explain how the web server (e.g., Flask or Django) connects the UI to the trained ML model.

API/Service Flow: Detail how user input is sent to the model and how predictions are displayed back to the user.

4.4 Testing

Include the testing used in your project.

Chapter 5

Results and Discussion

In Chapter 5, students provide evidence of the ”full functioning of working model as per stated requirements”. This chapter is critical for the ”Review 2” assessment, as it validates the ”implementation and testing of the solution”.

5.1 Working Prototype Demonstration

Screenshots and descriptions of the functional model. This section serves as the visual proof of the ”working model”.

Interface Walkthrough: Include high-quality screenshots of the web application’s frontend, including input forms and result displays.

System Functionality: Provide a step-by-step description of how a user interacts with the ML model through the web interface.

Feature Verification: Map each screenshot to the initial ”stated requirements” to prove the prototype is fully functional.

User Feedback (if any): Describe how the interface facilitates ”societal impact” by being accessible to the target users.

5.2 Validation and Inference

Drawing proper inferences from experimental results or simulations. Students must demonstrate their ability to ”draw the proper inferences from available results”.

Experimental Results: Present data obtained through ”theoretical, experimental, or simulations”.

Justification of Results: Provide a ”proper justification” for why the model produced specific outputs.

Inference Drawing: Explain what the results mean in the context of the ”societal/research needs” identified in Chapter 1.

Comparative Validation: Briefly discuss how these results compare to the ”proposed multiple solutions” considered during the design phase.

5.3 Performance metrics

Accuracy, cost-effectiveness, and efficiency analysis. In this section, we explain the performance metrics used to check the performance of model. These metrices include accuracy, precision, recall, and F1-score. This section quantifies the ”excellence” and ”efficiency” of the engineering solution.

ML Accuracy: Use standard metrics such as Accuracy, Precision, Recall, and F1-Score to evaluate the ”skill sets” applied in model training.

Confusion Matrix: Include a visual representation of the model's classification performance to support ”validation with proper justification”.

Cost-Effectiveness: Provide an analysis of ”cost effectiveness” by comparing the performance gained against the resources/computation used.

System Efficiency: Analyze the ”impact of solutions” in terms of response time and resource efficiency for ”sustainable development”.

Chapter 6

Conclusion and Future Work

In the final chapter, students summarize their findings and reflect on the project management principles and individual growth achieved during the semester. This chapter is essential for demonstrating the ”Course Outcomes” related to life-long learning and interpersonal skills.

6.1 Conclusion

This section provides a holistic summary of the ”working model” and its alignment with the initial project goals.

Project Summary: Briefly restate the problem and the specific Web-ML solution developed.

Requirement Validation: Explain how the final prototype meets all ”stated requirements” and addresses the identified ”societal/research needs”.

Impact Statement: Summarize the ”societal and environmental context” of the project, specifically how it contributes to the mapped Sustainable Development Goals (SDGs).

6.2 Future Scope:

This section demonstrates the ”capabilities of self-learning in a group, which leads to life-long learning”.

Technical Improvements: Suggest potential enhancements, such as using more advanced ML algorithms, improving UI/UX, or scaling the web infrastructure.

Feature Extension: Identify additional features that could be added to increase the project’s ”cost effectiveness and societal impact”.

Research Directions: Mention how this project could serve as a foundation for further research or a ”one-year project” extension in the next semester.

References

[1] A. Student and B. Guide, ”Web-based Machine Learning for Healthcare Monitoring,” International Journal of Computer Applications, vol. 175, no. 10, pp. 1-8, 2023.

[2] F. Pedregosa et al., ”Scikit-learn: Machine Learning in Python,” Journal of Machine Learning Research, vol. 12, pp. 2825-2830, 2011.

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